

HSC Investigating Science Stage 6 - Key Definitions

A Data-Driven Guide to the Full Syllabus

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Investigating Science Stage 6 — Definitions (Comprehensive Glossary)

Aligned to NSW Stage 6 Investigating Science. Concise, exam-ready definitions in Australian English. Use Cmd/Ctrl-F to search.

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A

- **Abstract:** A brief summary of the report’s purpose, key methods, principal results, and main conclusions.
- **Accuracy:** The closeness of measured values to the true or accepted value.
- **Account for:** State reasons for; report on and explain cause/effect.
- **Aim:** The overall purpose of an investigation, expressed as what it intends to determine or establish.
- **Analog technology:** Technology based on continuous signals, susceptible to noise and drift, often replaced by digital methods for precision and reproducibility.
- **Analyse:** Identify components and relationships; draw out and relate implications.
- **Anomaly:** An observation that deviates from the expected pattern and warrants investigation.
- **Apparatus:** Equipment and instruments used in an investigation.
- **Apply:** Use knowledge and understanding in a new situation.
- **Argument:** A claim supported by evidence and reasoning, addressing counterarguments where relevant.
- **Audience:** The intended readers or users of a communication product.

B

- **Bar chart:** A graph using bars to compare frequencies or values across categories.
- **Beneficence:** An ethical principle requiring benefits to outweigh risks; researchers should maximise benefits and minimise harms.
- **Bias:** Systematic deviation from the true value or an unbiased result due to design, selection, measurement, or interpretation.

- **Bias potential:** Aspects of study design or conduct that could introduce systematic error, such as non-random allocation, unblinded assessment, or selective reporting.
- **Biological plausibility:** Whether a proposed causal relationship is consistent with accepted biological mechanisms.
- **Blank:** A sample containing no analyte, used to correct for background signal.

C

- **Calculate:** Ascertain/determine from given facts, figures or information.
- **Calibration:** Adjustment or verification of an instrument against known standards to ensure accurate readings.
- **Calibration curve:** A plot of instrument response versus known standards used to determine unknown concentrations.
- **Capture–recapture:** A method for estimating population size by marking, releasing, and recapturing individuals.
- **Case study:** An in-depth investigation of a particular instance, event, or small number of cases.
- **Categorical data:** Data sorted into categories; may be nominal (unordered) or ordinal (ordered).
- **Causation:** A relationship in which a change in one variable directly produces a change in another, established through controlled design and elimination of confounders.
- **CER (Claim–Evidence–Reasoning):** A structured explanation linking a claim to evidence with scientific reasoning.
- **Cherry-picking:** Selecting only data that support a desired conclusion while ignoring contrary evidence.
- **Children:** A vulnerable population requiring additional ethical safeguards, including assent and parental consent.
- **Citation:** A formal reference to a source used; enables verification and avoids plagiarism.
- **Claim:** A clear, testable statement or conclusion that answers the research question.
- **Clarify:** Make clear or plain.
- **Classify:** Arrange or include in classes or categories.
- **Comparative study:** A design that compares groups or conditions without full control of variables.
- **Competence:** In ethics, the capacity of participants to understand information and make an informed decision.
- **Confidentiality:** Protecting participants' information from unauthorised disclosure.
- **Confidence:** The stated probability that a confidence interval contains the true value (e.g., 95% confidence).
- **Conflict disclosure:** A transparent statement of any interests that could influence research conduct or reporting.
- **Conflict of interest:** A situation where personal or financial interests could unduly influence judgements or reporting.

- **Confounding variable:** A factor associated with both the independent and dependent variables that can distort the estimated effect.
- **Consensus:** General agreement among experts based on convergent evidence.
- **Consequence:** The severity of outcome if a hazardous event occurs.
- **Construct:** Make; build; put together items or arguments.
- **Content analysis:** A systematic method for coding and interpreting qualitative texts, media, or transcripts.
- **Control group:** The group that does not receive the active treatment and is used for valid comparison.
- **Control measure:** An action that eliminates or reduces risk (e.g., engineering controls, procedures, PPE).
- **Controlled experiment:** An experiment that includes a control group or condition for valid comparison.
- **Correlation:** A statistical association between variables indicating they change together without implying causation.
- **Critically analyse:** Add a degree of accuracy, depth, knowledge and understanding, logic, questioning, reflection and quality to analysis.
- **Critically evaluate:** Add a degree of accuracy, depth, knowledge and understanding, logic, questioning, reflection and quality to evaluating.
- **Cross-sectional study:** A study that examines different subjects at a single point in time.

D

- **Data provenance:** The documented origin and processing history of a dataset.
- **Data visualisation:** The graphical representation of data to reveal patterns, trends, and relationships.
- **Deduce:** Draw conclusions from provided information; reason logically.
- **Define:** State the meaning of a word, phrase, or concept.
- **Dependent variable:** The variable measured as the response to changes in the independent variable.
- **Descriptive statistics:** Numerical summaries (e.g., mean, median, standard deviation) that describe features of a dataset.
- **Digital technology:** Technology that uses discrete binary signals, enabling error checking, replication, and high precision.
- **Dilution:** The process of reducing solute concentration, often by adding solvent.
- **Discussion:** The section that interprets results, evaluates method and data quality, and relates findings to the hypothesis and literature.
- **Discrete data:** Quantities that take distinct, separate values (often counts).
- **Dose–response relationship:** A pattern where changes in exposure level are associated with proportional changes in effect, supporting causation.

- **Double-blind trial:** A study in which neither participants nor assessors know the assigned treatment, reducing bias.
- **Drift:** A slow, systematic change in an instrument's response over time.

E

- **Effect size:** A quantitative measure of the magnitude of a relationship or difference (e.g., Cohen's d , r).
- **Elimination (risk control):** Removing the hazard entirely; the most effective control in the hierarchy.
- **Environmental impact:** The effect of an activity on the biophysical environment.
- **Evaluate:** Make a judgement based on criteria; determine the value of.
- **Evaluation:** A critical appraisal of methods, data quality, and conclusions, noting strengths and limitations.
- **Evidence:** Data or observations that support or refute a claim or hypothesis.
- **Evidence quality:** The trustworthiness of data, influenced by design, bias control, precision, and reproducibility.
- **Examine:** Inquire into; investigate.
- **Experimental investigation:** A study in which one or more variables are manipulated to test a causal hypothesis.
- **Expertise gaps:** Mismatches between an author's claimed expertise and the topic, which can reduce credibility.
- **Explain:** Relate cause and effect; make the relationships between things evident.
- **Extract:** Choose relevant and/or appropriate details.

F

- **Fair test:** An investigation in which only the independent variable is changed while all other relevant variables are controlled.
- **Falsifiability:** The property of a claim that it can be disproven by evidence from a possible observation or experiment.
- **Field log:** A contemporaneous record of methods, conditions, and observations during field work.
- **Field study:** An investigation conducted in a natural setting rather than a laboratory.
- **Field work:** Scientific investigation conducted in natural settings rather than controlled laboratory conditions.
- **Fraud detection:** Processes to identify fabrication, falsification, or plagiarism in research.
- **Funding bias:** Systematic influence on research outcomes associated with the funder's interests.

G

- **Generalisation (external validity):** The extent to which results can apply to other populations, places, or times.
- **GIS:** Geographic Information Systems; software for capturing, analysing, and visualising spatial data.
- **GPS:** Global Positioning System; a satellite-based navigation system for determining location.
- **Gradient (slope):** The rate of change of the dependent variable with respect to the independent variable.
- **Gross error:** A large mistake (e.g., misreading scale, recording error) that typically invalidates a data point.

H

- **Hazard:** A source or situation with potential to cause harm.
- **Hierarchy of control:** An ordered framework of risk controls from most to least effective: eliminate, substitute, engineer, administrate, PPE.
- **Histogram:** A plot showing the distribution of continuous data using contiguous bins.
- **Hypothesis:** A testable prediction about the relationship between variables, derived from observations or existing theory.

I

- **Identify:** Recognise and name.
- **Informed consent:** Voluntary agreement to participate after being informed of the study's purpose, procedures, risks, and benefits.
- **Infrastructure (research):** Facilities, equipment, and platforms that enable data collection, storage, and collaboration.
- **Institutional Review Board (IRB):** A committee that reviews research proposals for ethical compliance; in Australia, an HREC.
- **Instrument resolution:** The smallest increment of a quantity that an instrument can display.
- **Intercept:** The value of the dependent variable when the independent variable is zero (where a line crosses the axis).
- **Interquartile range:** The spread of the middle 50% of data; Q3 minus Q1.
- **Interpret:** Draw meaning from.
- **Introduction:** The section of a report that states the background, aim, and hypothesis or research question.

J

- **Justification:** A defence of a decision or conclusion using appropriate evidence and principles.
- **Justice:** An ethical principle requiring fair recruitment, burden-sharing, and access to benefits among participants.

K

- **Key performance indicator (KPI):** A metric used to evaluate the success of a process or intervention against targets.

L

- **Law:** A concise statement or mathematical relationship describing consistent natural regularities.
- **Layout (figures/tables):** Conventions for labelling axes, units, captions, and avoiding distortion.
- **Likelihood:** The estimated probability that a hazardous event will occur.
- **Line graph:** A graph connecting data points with lines to show changes over continuous intervals.
- **Line of best fit:** A line that minimises the overall deviation between itself and the data points, representing the trend.
- **Literature review:** A critical summary and synthesis of published research on a defined topic.
- **Longitudinal study:** A study that follows the same subjects over an extended period to observe change.
- **Low-quality evidence:** Findings with substantial risk of bias, imprecision, inconsistency, or indirectness.

M

- **Mean:** The arithmetic average of a set of values.
- **Measurement uncertainty:** A quantified estimate of the range within which the true value is expected to lie, reflecting measurement limitations.
- **Median:** The middle value when data are ordered; splits the dataset into two halves.
- **Method:** The section detailing materials, variables, and procedures to allow replication.
- **Methodology:** The systematic, theoretical analysis of methods applied to a field, encompassing the underpinning rationale for chosen methods.
- **Mode:** The most frequently occurring value in a dataset.

- **Modality:** The medium and mode of communication used (e.g., written report, poster, oral presentation).
- **Model (computational):** A model implemented in software to simulate system behaviour.
- **Model (conceptual):** A representation using ideas or diagrams to explain a system's structure or function.
- **Model (mathematical):** A set of equations describing relationships between variables in a system.
- **Model (physical):** A tangible representation that demonstrates a system's features or behaviour.
- **Modelling study:** A study that develops and analyses models to explain or predict phenomena.

N

- **Nocebo:** A harmful or adverse effect arising from negative expectations rather than an active treatment.
- **Nominal data:** Categorical data without an intrinsic order (e.g., species names, colours).
- **Normalisation:** Rescaling data to a common scale to enable comparison.
- **Null hypothesis:** A default claim that there is no effect or no difference, tested against an alternative hypothesis.

O

- **Observation:** Information obtained through the senses or instruments during an investigation.
- **Observational study:** Research where phenomena are observed without manipulation, used when experimentation is impossible or unethical.
- **Open science:** Practices that increase transparency and access, including open data, open methods, and preprints.
- **Outlier:** A data point that lies far from other observations and may indicate variability, error, or a novel effect.

P

- **Paradigm:** A dominant framework of theories, methods, and standards that guides scientific practice.
- **Pattern:** A recurring arrangement or relationship in data.
- **Peer review:** The evaluation of scientific work by other experts before or after publication to check quality and credibility.

- **Percentage:** A proportion expressed per hundred.
- **Percentage error:** The absolute error expressed as a percentage of the true or accepted value.
- **Plagiarism:** Presenting another's ideas or words as one's own without proper acknowledgment.
- **Placebo:** An inert treatment used as a control to isolate expectancy effects under blinding.
- **Pilot study:** A small-scale preliminary study to test feasibility, methods, and logistics prior to a full investigation.
- **Population (study):** The complete set of individuals or items about which conclusions are to be drawn.
- **Precision:** The closeness of repeated measurements to each other, regardless of their accuracy.
- **Precautionary principle:** The approach that lack of full scientific certainty should not delay measures to prevent serious or irreversible harm.
- **Predict:** Suggest what may happen based on available information.
- **Prepared mind (serendipity):** The idea that chance favours researchers who are knowledgeable and attentive to anomalies.
- **Primary data:** Data collected first-hand by the investigator for the specific purpose.
- **Procedure:** The ordered steps used to perform an investigation or part of it.
- **Processed data:** Data that have been cleaned, organised, or transformed from raw form.
- **Proportion:** A ratio expressing a part relative to the whole.
- **Propose:** Put forward (e.g., a point of view, idea, argument, suggestion) for consideration or action.
- **Protocol:** A detailed, standardised set of steps that specifies how to carry out a procedure.
- **Proxy measure:** An indirect measure that stands in for a variable that is difficult to measure directly.
- **Publish or perish:** Pressure to publish frequently, which can distort incentives and research practices.
- **Purpose:** The intended effect or outcome of the communication.

Q

- **Quadrat:** A square frame used to define a sample area for ecological surveys.
- **Qualitative data:** Non-numeric information describing qualities or categories (e.g., colour, type).
- **Quantitative data:** Numeric measurements that can be analysed statistically.

R

- **Random error:** Unpredictable variations in measurements that affect precision but can be minimised through repetition.
- **Randomisation:** Assigning treatments or selecting samples by chance to reduce selection bias and confounding.
- **Ratio:** A comparison of two quantities by division.
- **Raw data:** Original, unprocessed measurements recorded during an investigation.
- **Reasoning:** The logical process that connects evidence to a claim via scientific principles.
- **Reagent:** A substance used to cause a chemical reaction or detect the presence of another substance.
- **References:** The list of sources cited in the report, formatted consistently.
- **Reliable experiment:** Measuring reaction rates with multiple trials, calculating mean and standard deviation.
- **Reliability:** The consistency and reproducibility of experimental results, improved by replication, adequate sample size, and standardised procedures.
- **Reliability of sources:** The extent to which a source consistently provides accurate and trustworthy information.
- **Replicability:** The ability of an independent team to obtain consistent results using the same methods and data.
- **Replication:** Conducting an independent repeat of an entire study to test the robustness of findings.
- **Replicates:** Parallel, equivalent experimental units or measurements within the same study used to estimate variability.
- **Resolution:** The smallest increment of a quantity that an instrument can display.
- **Results:** The section presenting processed data, tables, and graphs without interpretation.
- **Retraction:** A formal withdrawal of a published paper due to serious error, misconduct, or irreproducibility.
- **Revision:** The modification of methods, models, or conclusions in response to new evidence.
- **Risk:** The combination of the likelihood and consequence of harm occurring.
- **Risk assessment:** A systematic process to identify hazards, evaluate risk, and select control measures.

S

- **Safety Data Sheet (SDS):** A document that provides information on the properties of a chemical and advice on safe handling.
- **Sample size:** The number of observations or experimental units; larger samples reduce random error and increase precision.

- **Sampling method:** The strategy used to select units from a population (e.g., random, stratified, systematic).
- **Scatter plot:** A graph showing individual data points for two variables to visualise relationships.
- **Selection bias:** Distortion arising when participants or samples are not representative due to the selection process.
- **Sensitivity:** The smallest change in the measurand that an instrument can detect reliably.
- **Serendipity in science:** Accidental discoveries that occur when investigating unrelated phenomena, requiring a prepared mind to recognise their significance.
- **Serial dilution:** A stepwise set of dilutions where each solution is diluted by the same factor.
- **Significant figures:** Digits in a number that convey precision, including all certain digits and the first uncertain digit.
- **Simulation:** An imitation of a real-world process or system to explore behaviour under varied conditions.
- **Single-blind trial:** A study in which participants do not know which treatment they receive, reducing expectancy effects.
- **Standard curve:** Another term for calibration curve.
- **Standard deviation:** A measure of the dispersion of data around the mean.
- **Standard operating procedure (SOP):** An approved, detailed instruction set that prescribes the exact method to perform a routine task for consistency and safety.
- **Standard solution:** A solution of accurately known concentration used for calibration or titration.
- **Stakeholder:** Any person or group affected by, or with an interest in, a scientific issue or project.
- **Sustainability:** Meeting present needs without compromising the ability of future generations to meet theirs.
- **Systematic error:** Consistent bias in measurements that affects accuracy and cannot be eliminated through repetition.

T

- **Theory:** A well-substantiated explanation of aspects of the natural world that integrates laws, hypotheses, and evidence.
- **Tolerance:** The allowable deviation from a specified value, usually set by design or manufacturer.
- **Transect:** A straight-line path along which samples or observations are made in field studies.
- **Transformed data:** Data converted mathematically (e.g., log, square root) to meet analysis assumptions.

- **Treatment:** A defined level or condition of the independent variable applied to experimental units.
- **Trend:** A consistent directional change in data over time or across levels of a variable.

U

- **Uncertainty:** A quantified estimate of the range within which the true value is expected to lie, reflecting measurement limitations.

V

- **Validity (external):** The extent to which results can be generalised beyond the study conditions to other settings or populations.
- **Validity (internal):** The degree to which an investigation establishes a trustworthy cause–effect relationship, free from confounding.
- **Validity of sources:** The degree to which a source’s content is sound, relevant, and supported by evidence.
- **Variable:** Any factor that can change or be controlled in an investigation.

W

- **Working hypothesis:** The current best explanation guiding data collection and analysis, subject to revision.

Y

- **Young persons (ethics):** Individuals who may have reduced capacity to consent; research requires assent and guardian consent where appropriate.

Research communication & NESAs “Glossary of Key Words” (exam verbs)

- **Account for:** State reasons for; report on and explain cause/effect.
- **Analyse:** Identify components and relationships; draw out and relate implications.
- **Apply:** Use knowledge and understanding in a new situation.
- **Assess:** Make a judgement of value, quality, outcomes or results.
- **Calculate:** Ascertain/determine from given facts, figures or information.
- **Clarify:** Make clear or plain.

- **Classify:** Arrange or include in classes or categories.
- **Compare:** Show how things are similar and different.
- **Construct:** Make; build; put together items or arguments.
- **Contrast:** Show how things are different or opposite.
- **Critically analyse:** Add a degree of accuracy, depth, knowledge and understanding, logic, questioning, reflection and quality to analysis.
- **Critically evaluate:** Add a degree of accuracy, depth, knowledge and understanding, logic, questioning, reflection and quality to evaluating.
- **Deduce:** Draw conclusions from provided information; reason logically.
- **Define:** State the meaning of a word, phrase, or concept.
- **Demonstrate:** Show by example; give a practical exhibition or explanation.
- **Describe:** Provide characteristics and features.
- **Discuss:** Identify issues and provide points for and/or against.
- **Distinguish:** Recognise or note/indicate as being distinct or different.
- **Evaluate:** Make a judgement based on criteria; determine the value of.
- **Examine:** Inquire into; investigate.
- **Explain:** Relate cause and effect; make the relationships between things evident.
- **Extract:** Choose relevant and/or appropriate details.
- **Identify:** Recognise and name.
- **Interpret:** Draw meaning from.
- **Investigate:** Plan, inquire into, and draw conclusions about.
- **Justify:** Support an argument or conclusion with evidence and reasoning.
- **Outline:** Sketch in general terms; indicate the main features.
- **Predict:** Suggest what may happen based on available information.
- **Propose:** Put forward (for example a point of view, idea, argument, suggestion) for consideration or action.
- **Recall:** Present remembered ideas, facts or experiences.
- **Recommend:** Provide reasons in favour; suggest a course of action.
- **Sketch:** Make a brief, general account or rough diagram; outline key features.
- **Summarise:** Express, concisely, the relevant details.
- **Synthesise:** Combine different ideas to create a new understanding.

If you want printable flashcards or a filtered set (e.g., only Measurement & Error), say “flashcards: [topic]” and I’ll add them.